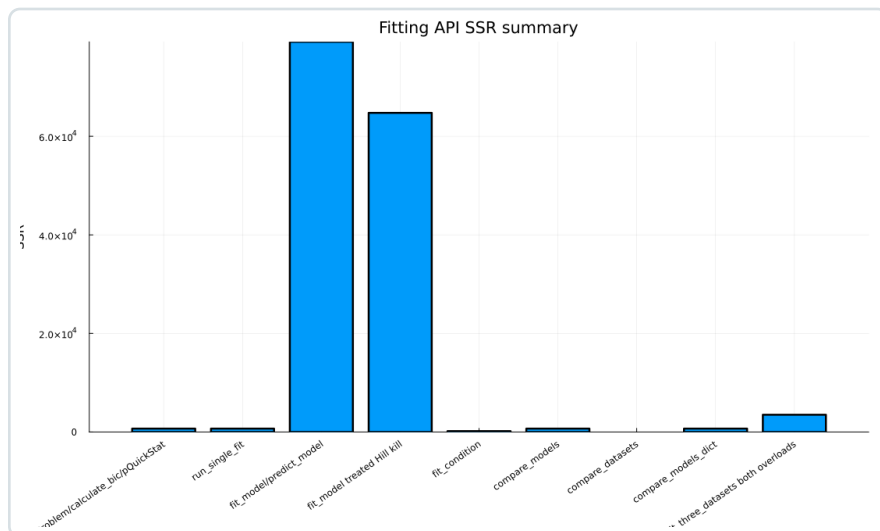


Fitting varieties

A practical guide to choosing, calling, and interpreting the package's fitting functions.

- One trajectory: `fit_model` or `fit_condition`.
- Competing model forms: `compare_models` or `compare_models_dict`.
- Independent experiments: `compare_datasets`.
- Shared multi-state and multi-condition biology: `run_joint_fit` plus `multistart` and `identifiability` tools.



FIT_MODEL

Fit one registered model to one time series

Best for a focused, single-population experiment when the model is available in the registry.

```
fit_model(model_spec::Registry.ModelSpec,  
          x::Vector{Float64}, y::Vector{Float64}, dose=0.0;  
          optimizer_method::Symbol=:de_rand_1_bin,  
          maxiters::Int=50_000, p0::Union{Vector{Float64},Nothing}=nothing,  
          anchor_params::Dict{Int,Float64}=Dict{Int,Float64}())
```

- x and y must be equal-length Float64 vectors; observations are sorted by time.
- model_spec supplies bounds, default solver, parameter names, and fixed-parameter rules.
- anchor_params fixes selected parameter indices to upstream values for staged fitting.
- Returns params, BIC, SSR, and solver return code. Check all four before reporting a fit.

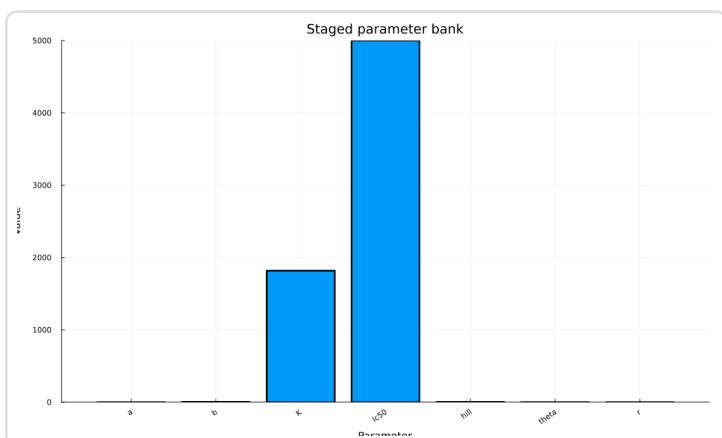
FIT_CONDITION

Fit one condition while retaining its DataFrame context

Use this after `build_conditions` when a condition contains all records for one biological scenario.

```
fit_condition(condition::FitCondition, model_spec::Registry.ModelSpec;  
  optimizer_method::Symbol=:de_rand_1_bin,  
  maxiters::Int=50_000, anchor_params::Dict{Int,Float64}=Dict())
```

- The condition carries time, count, dose, and metadata derived from the canonical long-format DataFrame.
- Use it in staged workflows where untreated monoculture estimates seed treated or coculture models.
- Keep the source table's error and replicate fields for plotting and later uncertainty analysis.

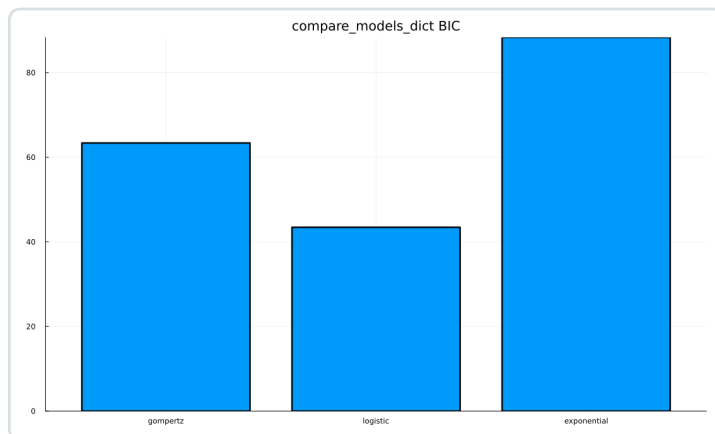


COMPARE_MODELS

Choose between two explicit candidate functions

Use when a clear scientific question has two alternatives, for example logistic versus Gompertz growth.

```
compare_models(x::Vector{<:Real}, y::Vector{<:Real},
               name1::String, model1::Function, p0_1::Vector{<:Real},
               name2::String, model2::Function, p0_2::Vector{<:Real};
               bounds1=nothing, bounds2=nothing, output_csv="model_comparison.csv")
```



- Each function must obey the ODE RHS convention (du, u, p, t).
- BIC balances fit quality against parameter count; lower is preferred among models that pass diagnostics.
- Use output_csv deliberately so a presentation can cite the exact comparison table.

COMPARE_DATASETS

Fit and summarize two experimental datasets

Useful for untreated versus treated, sensitive versus resistant, or one dataset per laboratory condition.

```
compare_datasets(x1::Vector{<:Real}, y1::Vector{<:Real},  
                name1::String, model1::Function, p0_1::Vector{<:Real},  
                x2::Vector{<:Real}, y2::Vector{<:Real},  
                name2::String, model2::Function, p0_2::Vector{<:Real};  
                bounds1=nothing, bounds2=nothing)
```

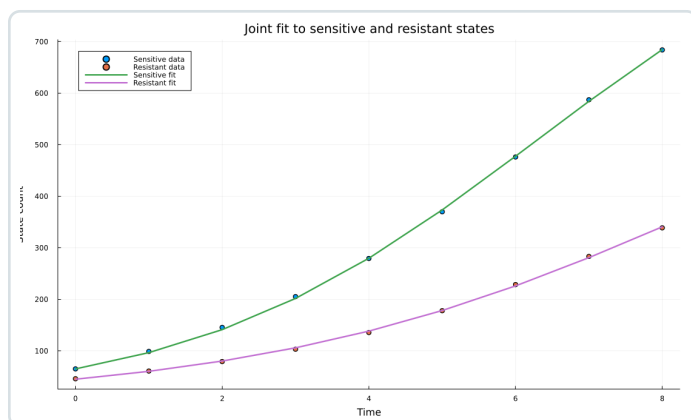
- This performs two fits and writes a compact dataset comparison table.
- It is not a shared-parameter likelihood. Use `run_joint_fit` when the two datasets must jointly constrain common parameters.
- Compare biological effect sizes only after each fit has passed residual and identifiability checks.

JOINT FITTING

Fit one parameter vector across multiple measured series

This is the workhorse for sensitive/resistant cocultures, damaged compartments, and shared drug-response parameters.

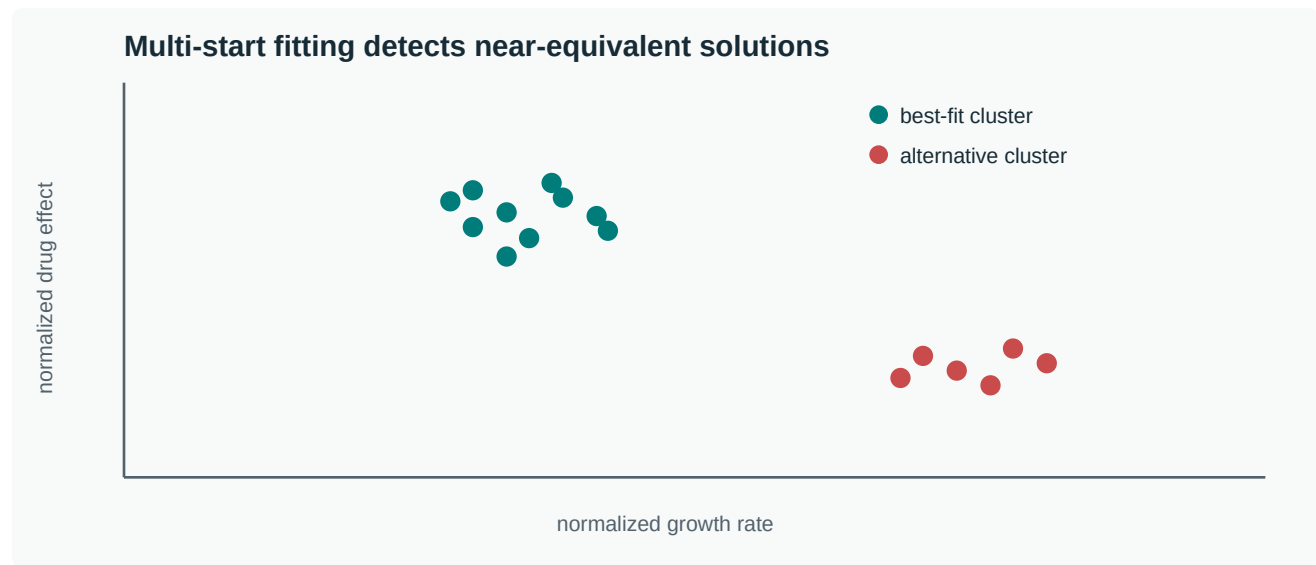
```
run_joint_fit(model::Function, dataset_specs::Vector{<:NamedTuple},
              u0::Vector{<:Real}, p0::Vector{<:Real};
              bounds=nothing, u0_builder=nothing, initial_time=nothing,
              optimizer::Symbol=:bfgs, maxiters::Integer=10_000)
```



- Each dataset spec needs x, y, and either state_index or observable.
- Set residual_scale per series so different assay variances are weighted appropriately.
- Use u0_builder for parameterized initial conditions and initial_time for day-zero seeding states.

A difficult fit deserves an audit trail, not one optimizer result

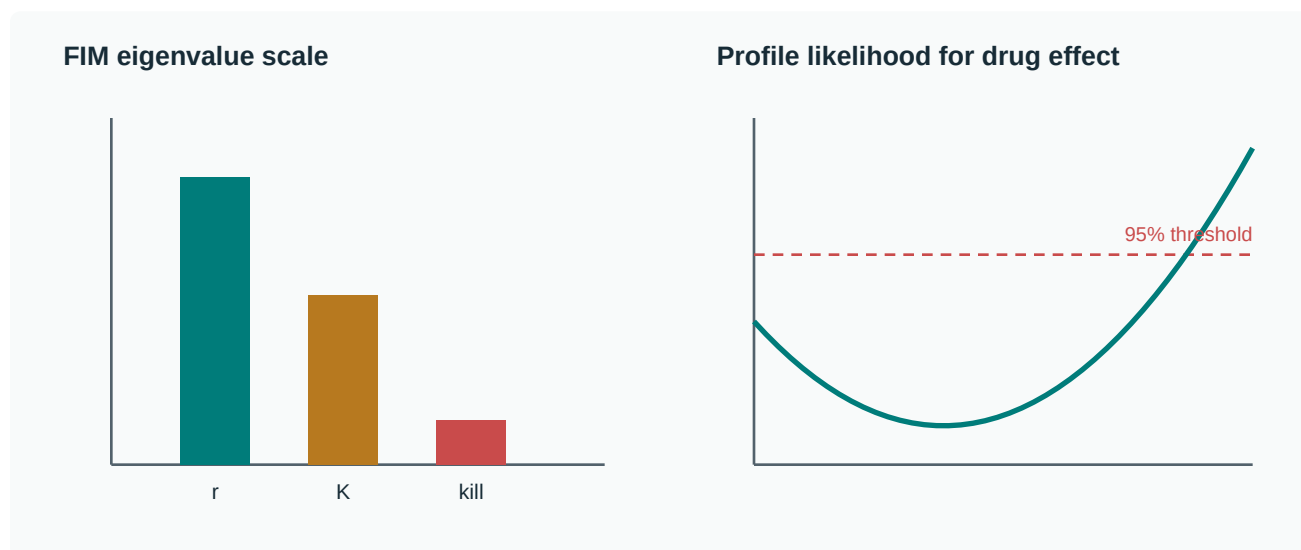
The new suite wraps joint fitting in checks that make parameter stability visible.



Function	Use
generate_multistarts	sample bounded initial guesses; use log scale for multi-order parameters
run_joint_multistart	retain best finite BIC and record every successful/failed start
practical_identifiability_report	combine clusters, FIM, profiles, and optional bootstrap gates

Three complementary tests for parameter reliability

These tools check different failure modes: local confounding, sampling variability, and planned-experiment recoverability.



- `profile_likelihood` fixes one parameter across a grid and refits the others; report whether the confidence region is bounded.
- `bootstrap_joint_fit` resamples residuals or simulates data using `residual_scale`, then refits the full joint model.
- `synthetic_recovery_benchmark` simulates the intended experiment at the actual time grid and asks whether known parameters are recovered.

STRUCTURAL ANALYSIS

Numerical stability is not a proof of global or local identifiability

Use the symbolic backend only after writing the exact model outputs and known inputs that match the data collection process.

```
map = ObservationMap("four_state", [:S, :D1, :D2, :M],
                    [:viable, :recoverable_damage, :terminal_damage, :macrophages])
validate_observation_map(map, dataset_specs)
result = structural_identifiability(symbolic_ode; mode=:global,
                                   prob_threshold=0.99)
```

- ObservationMap makes the assay-to-state mapping reviewable.
- structural_identifiability uses StructuralIdentifiability.jl and returns global, local, or nonidentifiable classifications.
- Known drug exposure belongs in the symbolic model; the outputs must match the fitted observables exactly.

DECISION GUIDE

Use the lightest function that answers the question

Escalate from simple fits to shared multi-state models only when experiment design and diagnostics justify the additional parameters.

Need	Use	Report with it
One curve	fit_model	params, BIC, SSR, overlay + error bars
Model hypothesis	compare_models	candidate list, BIC, residual checks
Two independent datasets	compare_datasets	per-dataset fit summaries
Shared biology across series	run_joint_fit	residual scales, state/observable map
Hard or high-dimensional model	multistart + identifiability report	clusters, FIM, profiles, bootstrap
Mechanistic uniqueness claim	structural_identifiability	symbolic ODE outputs and assumptions

Before fitting real data, validate the time grid, replicate structure, error model, and parameter bounds against a synthetic recovery benchmark.